

AI and Machine Learning Notes

1. Artificial Intelligence

Artificial Intelligence, or AI, is the field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include reasoning, learning, problem solving, and understanding language.

2. Machine Learning

Machine Learning is a branch of AI in which computers learn patterns from data. Instead of explicitly programming every rule, a machine learning model uses training data to learn relationships and make predictions on new data.

3. Deep Learning

Deep Learning is a subfield of machine learning that uses neural networks with multiple layers. Deep learning is commonly used for image recognition, speech recognition, natural language processing, and other complex tasks.

4. Embeddings

An embedding is a numerical representation of data such as text. Similar meanings are represented by vectors that are close to each other in vector space. Embeddings are widely used for semantic search and retrieval-augmented generation.

5. Vector Databases

A vector database stores numerical vector representations and allows efficient similarity search. ChromaDB is a vector database that can be used to store embeddings and retrieve documents that are semantically similar to a query.

6. Cosine Similarity

Cosine similarity measures how similar two vectors are based on the angle between them. It is commonly used in semantic search because vectors representing similar meanings tend to have similar directions.

7. Retrieval-Augmented Generation

Retrieval-Augmented Generation, or RAG, combines information retrieval with a language model. Relevant document chunks are retrieved from a vector database and provided as context to a language model. This allows the model to generate answers using information from the retrieved documents.

8. Ollama

Ollama allows developers to run large language models locally. A local model can receive retrieved document context and generate an answer without requiring the document content to be sent to an external service.